

hycfd: a general-purpose HDG hypersonic CFD driver

Design, conventions, solver machinery, and validation

myapps/hypersonic_cfd

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1 Overview and scope

hycfd is an implicit hybridizable discontinuous Galerkin (HDG) solver for the steady two-dimensional compressible Navier–Stokes equations, aimed at hypersonic perfect-gas flows and structured so that a two-temperature thermochemical-nonequilibrium model can be added behind the same interfaces. It was seeded from the verified **myapps/hdg_navierstokes** Exasim replica, which remains *frozen* as this code’s permanent regression oracle: every test run re-verifies that **hycfd** reproduces the frozen app’s Mach 8.03 half-cylinder wall data.

Capabilities: runtime polynomial order $p \geq 1$; triangle, quadrilateral, and mixed meshes; MFEM/Gmsh mesh-file input (curved elements supported); a boundary-condition catalog of seven types; dilatation-sensor shock capturing refreshed every Newton iteration; generic staged continuation (Mach/Reynolds homotopy and artificial-viscosity handovers through one mechanism); and MPI parallelism with a direct condensed-trace solve. Out of scope: three dimensions (assembly is dimension-parameterized but untested in 3D), time accuracy (pseudo-transient continuation only), and ablation/ALE coupling.

2 Governing equations and code units

The solver advances the conservative state $u = (\rho, \rho u_x, \rho u_y, \rho E)$ with the first-order system

$$q - \nabla u = 0, \quad (1)$$

$$\nabla \cdot (F^{\text{inv}}(u) - F^{\text{vis}}(u, q) - \varepsilon q) = 0, \quad (2)$$

where $\varepsilon(x)$ is the artificial viscosity (AV), applied as a Laplacian on *all* components. Note the convention $q = +\nabla u$ (Exasim uses $q = -\nabla u$; the sign map is confined to the generated kernels inside **physics/perfect_gas.hpp**).

Nondimensionalization (all derived at full precision from the typed **PerfectGasParams**, never from rounded deck literals):

- $\rho_\infty = 1$, $|u_\infty| = 1$; freestream pressure $p_\infty = 1/(\gamma M^2)$.
- Temperature is carried in internal-energy units: $T_{\text{code}} = e = p/((\gamma - 1)\rho)$, with freestream value $T_\infty^{\text{nd}} = 1/(\gamma(\gamma - 1)M^2)$ (**TinfND()**). The physical temperature enters only through Sutherland’s law via $T/T_\infty = \gamma M^2 p/\rho$ and the dimensional reference **T_inf_K**.
- The code viscosity includes the Reynolds division: $\mu_{\text{code}} = \mu_r(T)/\text{Re}$ with the Sutherland ratio $\mu_r(T) = (T/T_\infty)^{3/2}(T_\infty + S)/(T + S)$, $S = 110.4 \text{ K}$.

- Conductive heat flux = $f_c \nabla T_{\text{code}}$ with $f_c = \gamma \mu_{\text{code}} / \text{Pr}$.

Useful wall relations in these units (derived and verified against the flat-plate validation case, Section 9): for a boundary layer with freestream edge conditions,

$$c_f \sqrt{\text{Re}_x} = \sqrt{2} C_w f''(0), \quad q_w \sqrt{\text{Re}_x} = \frac{\gamma T_\infty^{\text{nd}} C_w \theta'(0)}{\text{Pr} \sqrt{2}}, \quad (3)$$

where $C_w = \rho_w \mu_w / (\rho_e \mu_e) = \mu_r(\theta_w) / \theta_w$ and $f''(0), \theta'(0)$ are the Lees–Dorodnitsyn similarity slopes.

3 HDG discretization

Volume unknowns (u, q) live in per-element L^2 spaces of order p ; the trace unknown $\hat{u}$ lives on mesh faces (`DG_Interface_FECollection`). The numerical flux is

$$\hat{F} \cdot n = F(\hat{u}, q) \cdot n + \tau (u - \hat{u}), \quad (4)$$

with a constant stabilization τ (deck key `physics.tau`; $\tau = 1$ is the correct scale here because $|u| + c \leq 1 + 1/M < 2$ in code units at any Mach number). Element assembly proceeds as: exact per-element gradient elimination $q = Cu - E\hat{u}$, elimination of u by a dense block solve, and static condensation onto $\hat{u}$. Mixed meshes are handled by per-geometry reference tables and per-element state offsets; quadrature order is $2p + \text{quadrature_increment}$ (default increment 1).

In parallel, elements are owned uniquely; the condensed local `SparseMatrix` is stitched across ranks by $P^T A P$ (`MakeSquareBlockDiag` + `ConvertFrom` + `MakePtAP`), which sums shared-face contributions. The condensed system is solved with MUMPS through PETSc (`input/petsc.opts`). Exasim-ordered reference data (`udg.bin`, `vdg.bin`) is mapped through explicit `serial_element_ids` because `ParMesh` rennumbers elements.

4 Physics interface and boundary conditions

All physics flows through the virtual `PhysicsModel` interface (runtime component count, flux with analytic Jacobians or an FD fallback, admissibility, wave speeds, named outputs) with `PerfectGasModel` as the sole current implementation. Boundary conditions are trace-row residuals $f_b(u, q, \hat{u}, n, x) = 0$ registered per model:

bc: type	Behavior
<code>freestream</code>	full-state Dirichlet at u_∞
<code>supersonic_outflow</code>	full extrapolation $u - \hat{u}$
<code>isothermal_wall</code>	no slip; energy pinned to T_{wall} (<code>Twall_K</code>)
<code>adiabatic_wall</code>	no slip; zero <i>numerical</i> energy flux $F_E \cdot n + \tau(u_E - \hat{u}_E) = 0$
<code>slip_wall</code>	$n \cdot \hat{u}_m = 0$, tangential extrapolation, adiabatic energy row
<code>characteristic_farfield</code>	flux splitting $A_n^+(u - \hat{u}) + A_n^-(u_\infty - \hat{u})$, frozen at freestream
<code>pressure_outlet</code>	mass/momentum extrapolated; energy row targets $E(p_{\text{out}})$

Two hard-won facts. First, the adiabatic condition must be imposed on the *numerical* flux: constraining only the conductive flux leaves $\hat{u}_E$ without a well-posed trace equation. Second, at high Reynolds number the adiabatic (and slip) wall energy level is a near-neutral mode with only an $O(1/\text{Re})$ conduction restoring force; cold starts against such walls are near-singular for steady Newton (see Section 10).

5 Nonlinear and linear solvers

DampedNewtonSolve (solvers/newton.hpp) implements damped Newton with two acceptance regimes:

- **Pure Newton:** Armijo backtracking, $|R(x + \alpha d)| \leq (1 - c_1 \alpha) |R(x)|$, halving from $\alpha = 1$ to `alpha_min`; failure aborts the solve.
- **PTC** (pseudo-transient continuation): backward-Euler volume mass term $M/\Delta\tau$, switched-evolution-relaxation step growth, and *plain-decrease* acceptance — the exactly-solved damped step reduces the residual by only $O(\Delta\tau)$, so the Armijo test would reject systematically. Rejection at every α shrinks $\Delta\tau$. PTC turns off below `newton.ptc_off_residual`.

Optional guard `require_admissible`: line-search trials with non-positive minimum density or pressure are rejected exactly like non-finite residuals. With the smoothed ρ/p floors active (`regularization: floors`), a globally decreasing step can otherwise park the state in floor-land, where the Jacobian is degenerate and Newton stalls.

The `NewtonPrepare` hook is called at the top of every iteration, before assembly, with the previous iteration’s residual; whatever it installs stays frozen through that iteration’s Jacobian and line search. It exists to support adaptive frozen-per-iteration artificial viscosity (next section) but is generic.

6 Shock capturing

The dilatation sensor (`discretization/av_sensor`) builds an elementwise amplitude

$$\varepsilon_K = \lambda h_K \max_{\text{qp}} \left[(|u| + c) s \left(\frac{-\nabla \cdot u h_K}{(p + 1) \kappa c} \right) \right], \quad (5)$$

with a compact-support smoothstep ramp s (identically zero below 0.2, one above 0.8) — resolved smooth flow produces *exactly* zero AV — followed by vertex-max smoothing onto a continuous H^1 order-1 field (parallel via `GroupCommunicator` max reduction).

`SensorAVController` realizes *adaptive frozen-per-iteration* AV: at the top of every Newton iteration the sensor is re-evaluated from the current iterate, under-relaxed by an exponential moving average (`av.relax`), floored by a bootstrap profile that decays linearly over `av.bootstrap.decay_iters` iterations (cold starts need AV where the shock *will* form), and installed as frozen tabulated data. Once the residual drops below `av.freeze_residual` (after the bootstrap window) the field freezes permanently and Newton closes quadratically. The per-solve lagged fixed point this replaced limit-cycled: the shock drifted between solves faster than outer under-relaxation could track. At Mach 8 the sensor–shock fixed point hovers chaotically around 10^{-3} without the freeze; with it, the campaign closes in one iteration from the freeze

point. Setting `freeze_residual = ∞` gives install-once frozen-sensor semantics (evaluate once from the initial state, never refresh).

Two interaction rules learned the hard way: frozen AV must never overlap a damped-initialization layer (it wrecks the first Newton directions), and switching AV profiles between warm-started stages is itself a shock to the solve — prefer install-once sensors or same-profile stages.

7 Continuation

`continuation.stages` is a list of YAML *override maps* applied on top of the base deck (not cumulatively). Allowed keys: `physics`, `av`, `newton`, `ptc`. Maps merge recursively, except that a map whose `mode` differs replaces the block wholesale (so stale sibling keys cannot leak across AV modes). The physics model, operator, and output writer are rebuilt per stage; the state carries over with the trace re-initialized from the interior. Mach/Reynolds homotopy, AV ramp-downs, and sensor-to-tanh handovers are all expressed through this one mechanism. The flagship campaign (`input/mach8_sensor.yaml`) runs a sensor stage from a damped freestream followed by three tanh ramp-down stages onto the frozen reference AV profile, and reproduces the frozen app’s wall data to 1.2×10^{-5} (pressure coefficient) and 2.4×10^{-5} (heat flux).

8 Input deck reference

Key	Meaning
<code>case_name</code>	run label for output collections
<code>mesh.source</code>	<code>exasim</code> <code>analytic</code> <code>file</code>
<code>mesh.exasim_dir</code>	Exasim run directory (regression only)
<code>mesh.file</code> , <code>mesh.curved_order</code>	MFEM/Gmsh mesh input
<code>mesh.nr</code> , <code>mesh.nc</code> , <code>mesh.order</code>	analytic half-cylinder annulus
<code>discretization.order</code>	polynomial order p (default 4)
<code>discretization.quadrature_increment</code>	polynomial order $2p + \text{inc}$
<code>bc:</code>	list of <code>{attrs: [...], type, params}</code> entries; every boundary attribute must be covered exactly once
<code>physics.model</code>	<code>perfect_gas</code>
<code>physics.gamma</code> , <code>reynolds</code> , <code>prandtl</code> , <code>mach</code>	nondimensional group
<code>physics.T_inf_K</code> , <code>T_wall_K</code>	Sutherland reference and wall temperature [K]
<code>physics.tau</code>	HDG stabilization
<code>physics.regularization</code>	<code>floors</code> enables the smoothed ρ/p floors (recommended for AV transients)
<code>av.mode</code>	<code>tanh</code> <code>sensor</code> <code>file</code>
<code>av.lambda</code> , <code>av.c</code>	tanh profile $\lambda \tanh(c(r - 1))$
<code>av.lambda</code> , <code>av.kappa</code>	sensor amplitude and ramp scale
<code>av.relax</code>	EMA weight on the fresh sensor field, $(0, 1]$

<code>av.freeze_residual</code>	permanent sensor freeze threshold (0 disables)
<code>av.bootstrap.{lambda,c,decay_iters}</code>	increasing tanh floor for cold starts
<code>init.mode</code>	<code>damped_freestream</code> (with c) <code>udg_file</code>
<code>newton.max_iters, tol</code>	iteration controls
<code>newton.armijo_c1, alpha_min</code>	line-search controls
<code>newton.ptc_off_residual</code>	PTC shutoff threshold
<code>ptc.enabled, initial_dt</code>	pseudo-transient continuation
<code>continuation.enabled,</code> <code>stages, history_csv,</code> <code>summary_csv</code>	staged overrides (Section above)
<code>linear.type</code>	<code>petsc_direct</code> (MUMPS)
<code>output.path, paraview_every</code>	ParaView output
<code>output.wall_csv,</code> <code>comparison.report</code>	wall data extraction
<code>comparison.*</code>	gated comparison against a reference wall CSV and shock standoff

9 Verification and validation

`make check` runs the fast gates: generated-kernel parity against the Exasim oracle headers (10^{-16}), FD Jacobian checks for every flux and boundary condition, manufactured-solution convergence at $p = 1..4$ on quadrilateral/triangle/mixed/curved meshes, the PTC limit fixture, and the sensor-AV unit gates. `make test` adds the $np = \{2, 4\}$ parallel consistency fixtures, the frozen Mach 8 regression (serial and $np = 4$; wall C_p within 8.8×10^{-6} , q_w within 2.1×10^{-5} of the frozen oracle), the M4.5 sanity and sensor decks, the Mach 8 sensor campaign, and the validation suite below, serial and at $np = 4$.

G7 validation suite (`tests/test_validation.cpp`)

Quantity	Reference	Result	Gate
Cylinder standoff, $M=3$	Sinclair–Cui Eq. (32)	+4.5%	8%
Cylinder standoff, $M=5$	Sinclair–Cui Eq. (32)	+6.6%	8%
Stagnation C_p , $M=3$	Rayleigh pitot	−2.3%	4%
Stagnation C_p , $M=5$	Rayleigh pitot	−3.1%	4%
Plate c_f , $M=4$, $\text{Re}=10^6$	compressible Blasius	0.75%	4%
Plate q_w , $M=4$, $\text{Re}=10^6$	compressible Blasius	0.024%	4%
Oblique shock p_2/p_1 , $M=3$, 10°	$\theta-\beta-M$	0.02%	2%
Oblique shock angle	$\theta-\beta-M$	1.8°	2.5°

Notes on the reference choices. The bow-shock standoff is gated against Sinclair & Cui, *Phys. Fluids* **29**, 026102 (2017), Eq. (32), not Billig’s correlation: Billig’s cylinder fit underpredicts

standoff for $M \lesssim 6$ (their Fig. 7, against Euler solutions and the Alperin/Kim/Kaattari experiments), which this suite reproduces — the computed standoffs sit 4–7% above Sinclair–Cui and 18%/9% above Billig. The standoff itself is measured by an outer-in density mid-jump crossing on the stagnation line (`StagnationDensityCrossing`); the older maximum-gradient diagnostic locks onto the near-wall thermal layer at high Mach and is retained only as a regression fingerprint. The stagnation C_p runs 2–3% below the inviscid pitot value because the AV-thickened shock produces slightly more entropy than a discontinuity.

The cylinder and plate cases are also available as driver decks for field exploration in ParaView (`input/m30.billig_cylinder.yaml`, `input/m50.billig_cylinder.yaml`, `input/m40.blasius_plate.yaml`) — build the plate mesh with `make input/plate_nx45_ny32.mesh` — the quantitative gates remain in the test. The oblique-shock case relies on the test-only Dirichlet override; passing `paraview` to `test_validation` saves every case’s converged solution to `ParaView/validation_*`.

The flat plate solves a Reynolds continuation ladder $\{10^5, 3 \times 10^5, 10^6\}$ at $M = 4$ (cold isothermal wall, pure Newton, admissibility guard, damped start at the resolved boundary-layer scale, plate beginning at the inflow plane) and compares wall c_f and q_w pointwise against a compressible-Blasius shooting solution (Sutherland $C(\theta)$, RK4 with two-parameter Newton shooting and Mach continuation; the solver self-checks the incompressible limit $c_f \sqrt{\text{Re}_x} = 0.664$). The oblique shock case prescribes the exact Rankine–Hugoniot piecewise weak solution on all boundaries through the Dirichlet override, seeds the interior with its projection, and verifies that the captured, sensor-regularized shock stays put and reproduces the downstream state.

10 Known limitations and practical guidance

- **Slip-wall cold starts.** A state with $O(1)$ residual concentrated in slip-wall trace rows (e.g. an impulsive freestream over a ramp) produces pathologically scaled Newton corrections ($|\delta| \sim 10^2$ for a residual of 10^{-2} , descent only at $\alpha \sim 10^{-3}$), independent of the initialization, Reynolds number, AV, and PTC. PTC cannot help: it damps only the volume block, and the condensed system degenerates toward the bare trace block as $\Delta\tau \rightarrow 0$ (it can reach exact MUMPS singularity). This is why the compression-corner validation is formulated as a Dirichlet-driven oblique shock. Candidate fixes: trace-block pseudo-time damping, or a rescaled slip-row formulation.
- **PTC on strongly anisotropic meshes** inherits the same $\Delta\tau \rightarrow 0$ degeneracy; the flat-plate case therefore runs pure Newton with the admissibility guard.
- **Leading-edge singularities** (sharp flat plate) develop a spurious expansion spike that deepens with Reynolds number; at $M = 6$ this defeats Newton beyond $\text{Re} \approx 10^5$ on the current meshes, and a frozen AV patch cannot be switched on mid-ladder without shocking the warm start. The validation case runs at $M = 4$, where the spike stays benign through $\text{Re} = 10^6$.
- **Sensor-AV runs are chaotic run-to-run** (element on/off flicker amplifies solver-level floating-point noise). Gates must not depend on exact transient trajectories; the residual freeze and tanh handover make campaign endpoints deterministic.
- **Pure-sensor solutions cannot match the frozen oracle’s wall heat flux:** the Exasim reference AV is the global $0.025 \tanh(5(r - 1))$ profile, which dominates the physical

viscosity through most of the boundary layer. Campaigns gated against the oracle must hand over to that profile first.